

Neural Network-Based Fault Detection

Architecture, Tuning, and Performance Evaluation

Seyed Rumaiz

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Chapter 1

Neural Network Model

1.1 Model Architecture

A feedforward Artificial Neural Network (ANN) was designed for multi-class fault detection. The network consists of three fully connected hidden layers using ReLU activation functions. Batch normalization and dropout regularization were applied after each hidden layer to improve training stability and generalization performance. The output layer employs a softmax activation function to produce class probability estimates.

- **Input Layer:** Dimension equal to the number of input features.
- **Hidden Layer 1:** 128 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Hidden Layer 2:** 64 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Hidden Layer 3:** 32 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Output Layer:** Softmax activation with four output neurons corresponding to fault classes.

Chapter 2

Training Procedure

2.1 Model Compilation

The neural network was compiled using the Adam optimizer and categorical cross-entropy loss, which is appropriate for one-hot encoded multi-class targets. Model performance was monitored using accuracy, precision, recall, and multi-label AUC metrics.

2.2 Training Strategy

The network was trained for a maximum of 20 epochs with a batch size of 32. Early stopping was applied based on validation AUC with a patience of 10 epochs. Class weights were used due to severe class imbalance.

Chapter 3

Base ANN Results

3.1 Classification Report

Table 3.1: Classification Report for Base ANN

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	232,587
1	1.00	1.00	1.00	1,205
2	0.98	0.98	0.98	37,694
3	0.98	1.00	0.99	1,200
Accuracy	0.99			
Macro Avg	0.99	0.99	0.99	272,686
Weighted Avg	0.99	0.99	0.99	272,686

3.2 Confusion Matrix

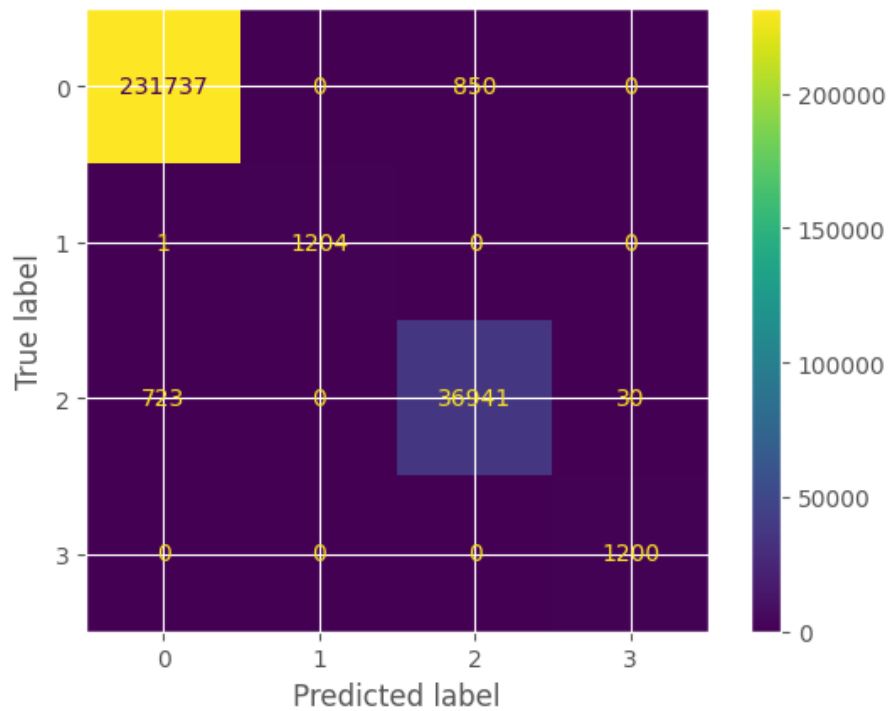


Figure 3.1: Confusion matrix for Base ANN.

3.3 Training and Validation Curves

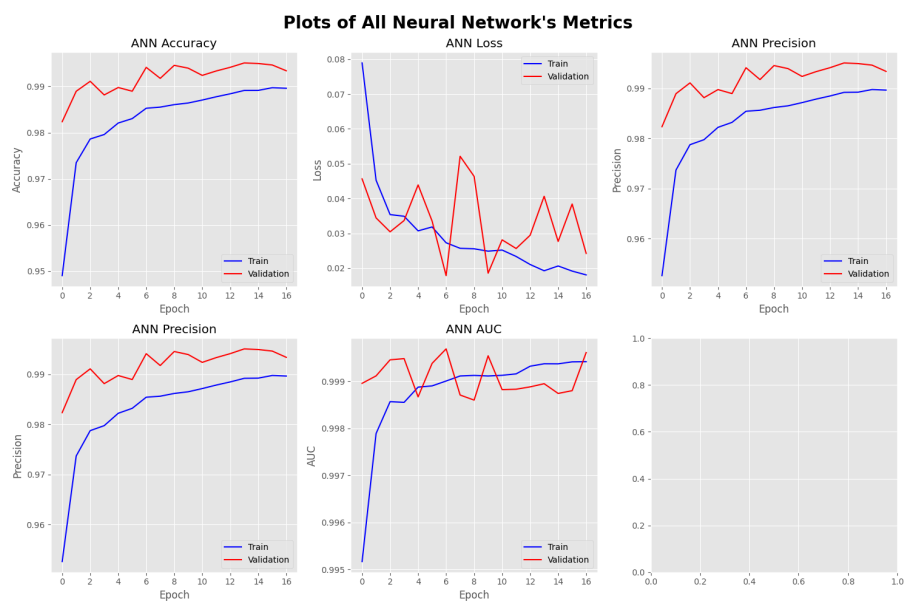


Figure 3.2: Training and validation curves for Base ANN.

3.4 Test Metrics

Table 3.2: Overall Test Metrics for Base ANN

Metric	Value
Test Loss	0.018
Test Accuracy	0.994
Test Precision	0.994
Test Recall	0.994
Test AUC	1.000

Chapter 4

Best / Tuned ANN Results

4.1 Classification Report

Table 4.1: Classification Report for Best ANN

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	232,587
1	1.00	1.00	1.00	1,205
2	0.98	0.99	0.98	37,694
3	0.98	1.00	0.99	1,200
Accuracy	1.00			
Macro Avg	0.99	1.00	0.99	272,686
Weighted Avg	1.00	1.00	1.00	272,686

4.2 Confusion Matrix

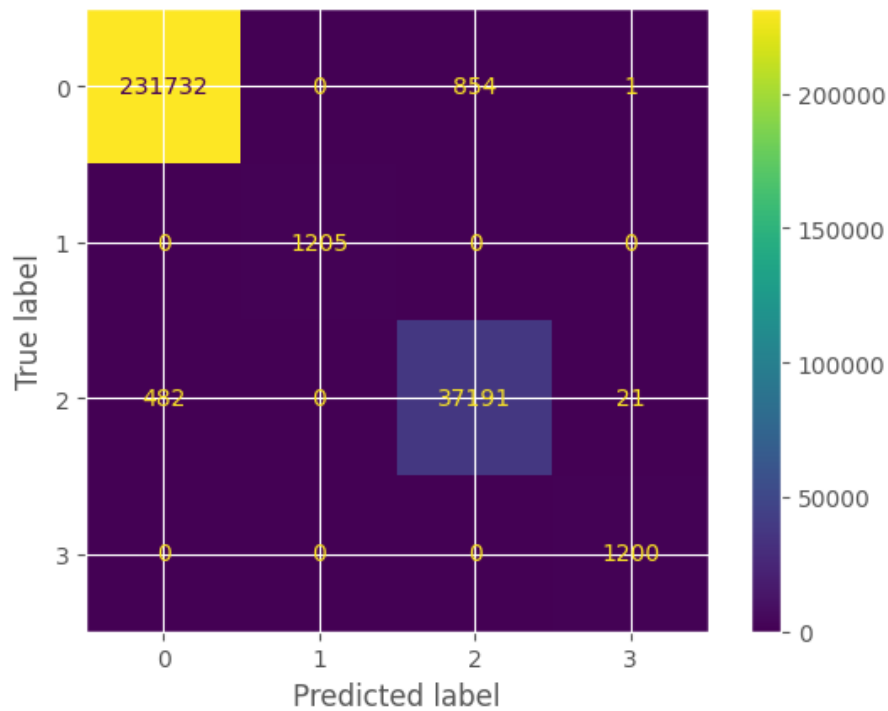


Figure 4.1: Confusion matrix for Best ANN.

4.3 Training and Validation Curves

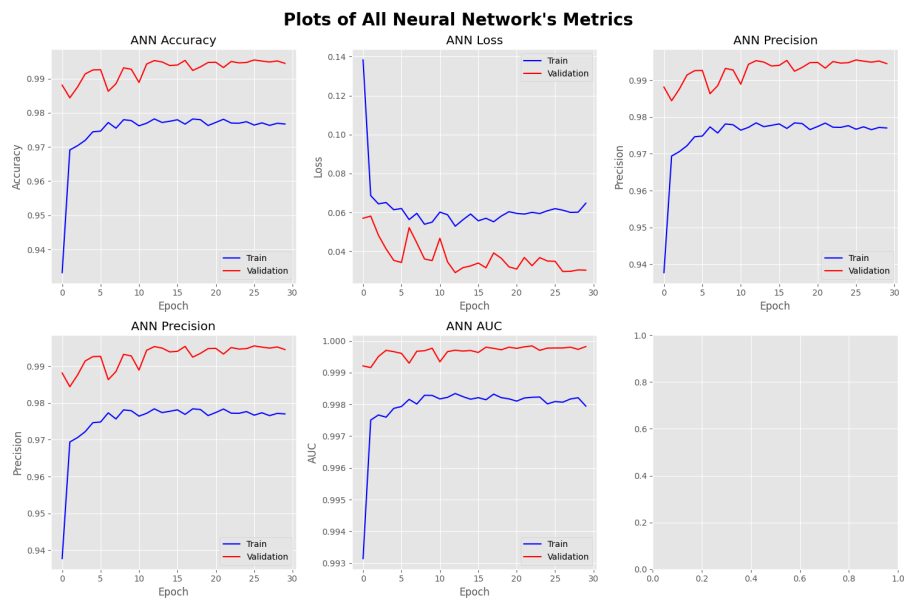


Figure 4.2: Training and validation curves for Best ANN.

4.4 Test Metrics

Table 4.2: Overall Test Metrics for Best ANN

Metric	Value
Test Loss	0.033
Test Accuracy	0.995
Test Precision	0.995
Test Recall	0.995
Test AUC	1.000

4.5 Best Hyperparameters

Table 4.3: Hyperparameters for Best ANN

Hyperparameter	Value
Number of layers	2
Units in Layer 0	128
Units in Layer 1	96
Units in Layer 2	96
Dropout rate	0.4
L2 regularization	0.000429
Learning rate	0.000508

4.6 Training Summary

The hyperparameter tuning was performed across 5 trials. The total elapsed time for all trials was approximately 03h 05m 26s.

Chapter 5

Explainable AI (SHAP) Visualizations

5.1 SHAP Summary Plot

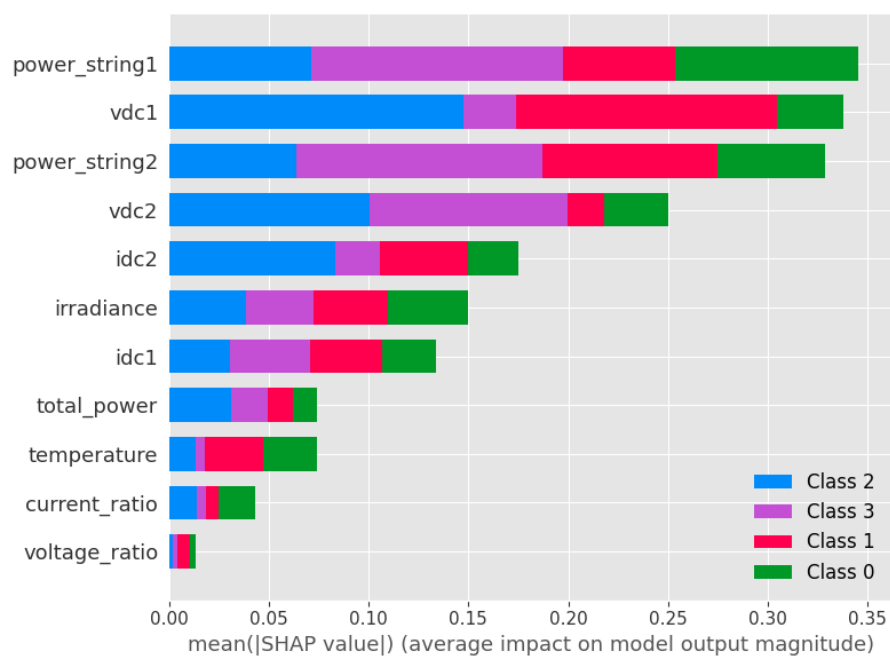


Figure 5.1: SHAP summary plot for the Best ANN.

5.2 SHAP Force Plot



Figure 5.2: SHAP force plot for the Best ANN.

Chapter 6

Conclusion

In this study, a neural network was designed and evaluated for multi-class fault detection. The Base ANN performed very well overall, but the Best/Tuned ANN showed slightly better performance on the minority classes due to careful hyperparameter tuning, including dropout, layer sizes, learning rate, and L2 regularization.

Regularization and early stopping helped prevent overfitting, and the SHAP analyses provided clear insights into which features influenced predictions the most.

Overall, the Best ANN offers highly accurate and reliable fault detection, while also giving interpretable results, making it suitable for practical deployment in real-world systems.